Lecture 8: Modelling Proportions — Logistic Regression, Overdispersion and Odds Ratios

MATH3701 Statistical Modelling

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Reading

Course notes: Chapter 5, Sections 5.1–5.4

`MATH3701 Statistical Modelling notes, Chapter 5`

Binary and Binomial Responses

Bernoulli trials:

$$B = \begin{cases} 1 & \text{if event occurs (“success”)} \\ 0 & \text{otherwise (“failure”)} \end{cases}$$

with $\mathbb{P}(B = 1) = p$.

Binomial distribution:
Sum of m independent Bernoulli trials with same p:

$$Y = \sum_{j=1}^{m} B_j \sim \text{Binomial}(m, p)$$

Special case: $m = 1$ gives Bernoulli (binary) data.

The Logistic Regression Model

Model specification:

$$Y_i \sim \text{Binomial}(m_i, p_i)$$

with the logit link:

$$\text{logit}(p_i) = \log \frac{p_i}{1 - p_i} = \mathbf{x}_i' \boldsymbol{\beta}$$

Equivalently:

$$p_i = \frac{\exp(\mathbf{x}_i' \boldsymbol{\beta})}{1 + \exp(\mathbf{x}_i' \boldsymbol{\beta})} = \frac{1}{1 + \exp(-\mathbf{x}_i' \boldsymbol{\beta})}$$

This ensures $0 < p_i < 1$ for all $\boldsymbol{\beta}$.

Interpreting Coefficients: Odds

Odds of success:

$$\text{Odds} = \frac{p}{1-p}$$

Example:

- $p = 0.8$: Odds $= 0.8/0.2 = 4$ ("4 to 1")
- $p = 0.5$: Odds $= 1$ ("even odds")
- $p = 0.2$: Odds $= 0.25$ ("1 to 4")

The logit is the log-odds:

$$\text{logit}(p) = \log(\text{Odds})$$

Interpreting Coefficients: Odds Ratios

Model: $\text{logit}(p) = \alpha + \beta x$

For a unit increase in x:

$$\begin{aligned} \text{logit}(p_{x+1}) - \text{logit}(p_x) &= \beta \\ \log \frac{\text{Odds}_{x+1}}{\text{Odds}_x} &= \beta \\ \frac{\text{Odds}_{x+1}}{\text{Odds}_x} &= e^{\beta} \end{aligned}$$

Interpretation:

$$\boxed{e^{\beta} = \text{Odds Ratio}}$$

A unit increase in x multiplies the odds by e^{β}.

Odds Ratio Examples

β	e^{β}	Interpretation
0	1.00	No effect
0.5	1.65	Odds increase by 65%
1.0	2.72	Odds nearly triple
-0.5	0.61	Odds decrease by 39%
-1.0	0.37	Odds reduced to 37%

Key point:

- $e^{\beta} > 1$: Higher $x \Rightarrow$ higher probability of success
- $e^{\beta} < 1$: Higher $x \Rightarrow$ lower probability of success
- $e^{\beta} = 1$ ($\beta = 0$): x has no effect

Maximum Likelihood Estimation

Log-likelihood:

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^{n} \left\{ y_i \log p_i + (m_i - y_i) \log(1 - p_i) + \log \binom{m_i}{y_i} \right\}$$

where $p_i = \text{logit}^{-1}(\mathbf{x}_i'\boldsymbol{\beta})$.

No closed form solution!
Solved iteratively using Fisher Scoring / IRLS.

Fitted values:

$$\hat{p}_i = \frac{\exp(\mathbf{x}_i'\hat{\boldsymbol{\beta}})}{1 + \exp(\mathbf{x}_i'\hat{\boldsymbol{\beta}})}, \qquad \hat{y}_i = m_i \hat{p}_i$$

Residuals for Binomial GLMs

Pearson residuals:

$$e_i^P = \frac{y_i - m_i\hat{p}_i}{\sqrt{m_i\hat{p}_i(1-\hat{p}_i)}}$$

Deviance residuals:

$$e_i^D = \text{sign}(y_i - m_i\hat{p}_i)\sqrt{d_i}$$

where

$$d_i = 2\left\{ y_i \log\frac{y_i}{m_i\hat{p}_i} + (m_i - y_i)\log\frac{m_i - y_i}{m_i(1-\hat{p}_i)} \right\}$$

For large m_i, residuals should be approximately $\mathcal{N}(0,1)$.

Deviance for Logistic Regression

Model deviance:

$$D = 2\sum_{i=1}^{n}\left\{y_i \log\frac{y_i}{\hat{y}_i} + (m_i - y_i)\log\frac{m_i - y_i}{m_i - \hat{y}_i}\right\}$$

Goodness of fit: Under correct model, $D \sim \chi^2_{n-r}$.

Alternative: Pearson χ^2 statistic

$$X^2 = \sum_{i=1}^{n}\frac{(y_i - m_i\hat{p}_i)^2}{m_i\hat{p}_i(1-\hat{p}_i)}$$

Both D and X^2 are asymptotically χ^2_{n-r}.

Fitting Logistic Regression in R

For grouped binomial data:

```
# y = number of successes, m = number of trials
y <- cbind(successes, failures)
model <- glm(y ~ x1 + x2, family = binomial)

# Or equivalently
model <- glm(cbind(successes, total - successes) ~ x1 + x2,
             family = binomial)
```

For binary (0/1) data:

```
# y is 0 or 1 for each observation
model <- glm(y ~ x1 + x2, family = binomial)
```

Example: R Output

```
Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -60.7175     5.1805  -11.72   <2e-16 ***
dose         34.2703     2.9122   11.77   <2e-16 ***

    Null deviance: 284.202  on 7  degrees of freedom
Residual deviance:  11.232  on 6  degrees of freedom
AIC: 41.43
```

Interpretation:

- $\hat{\beta} = 34.27$: log-odds ratio for unit dose increase
- $e^{34.27}$: odds ratio (very large)
- Residual deviance: 11.23 on 6 df — reasonable fit

Analysis of Deviance for the Beetle Data

```
anova(glm.fit)
```

```
Terms added sequentially (first to last)

     Df Deviance Resid. Df Resid. Dev  Pr(>Chi)
NULL                     7    284.202
dose  1   272.97         6     11.232  < 2.2e-16 ***
```

Adding `dose` reduces the deviance by 272.97 on 1 df — an overwhelming likelihood-ratio test for the effect of dose.

Hypothesis Testing

Three types of tests:

1. **Wald test** (from summary output):

$$z = \frac{\hat{\beta}_j}{\text{SE}(\hat{\beta}_j)} \sim \mathcal{N}(0,1) \text{ under } H_0 : \beta_j = 0$$

2. **Likelihood ratio test** (deviance difference):

$$D_0 - D_1 \sim \chi^2_{r_1 - r_0} \text{ under } H_0$$

3. **Goodness-of-fit test:**

$$D \sim \chi^2_{n-r} \text{ under correct model}$$

Summary: Logistic Regression

Key points:

- Logistic regression models binomial/binary responses
- Logit link: $\text{logit}(p) = \log \frac{p}{1-p} = \mathbf{x}'\boldsymbol{\beta}$
- Coefficients are log-odds ratios
- e^{β_j} = multiplicative effect on odds per unit increase in x_j
- MLE via iterative methods (Fisher Scoring)
- Deviance and Pearson χ^2 for goodness of fit
- Use `family = binomial` in R

Overdispersion: The Problem

Binomial assumption:

$$\text{Var}[Y_i] = m_i p_i (1 - p_i)$$

In practice: The observed variance often exceeds this.

Detection:

- Residual deviance $\gg$ residual degrees of freedom
- Ratio $D/(n - r) \gg 1$

Example: $D = 45.2$ on 10 df suggests overdispersion.

Causes of Overdispersion

Possible sources:

1. **Missing explanatory variables:**
   - Important variables not in the model
   - Unmeasured heterogeneity
2. **Incorrect link function:**
   - Logit may not be appropriate
3. **Lack of independence:**
   - Trials within groups may be correlated
   - Clustering effects

2×2 Contingency Tables

Setup:

	Success	Failure	Total
Group 1	y_1	$m_1 - y_1$	m_1
Group 2	y_2	$m_2 - y_2$	m_2

Probabilities:

- Group 1: $\mathbb{P}(\text{success}) = \pi_1$
- Group 2: $\mathbb{P}(\text{success}) = \pi_2$

Question: Is there an association between group and outcome?

Odds and Odds Ratio

Odds of success in each group:

$$O_1 = \frac{\pi_1}{1-\pi_1}, \qquad O_2 = \frac{\pi_2}{1-\pi_2}$$

Odds ratio:

$$\boxed{\psi = \frac{O_1}{O_2} = \frac{\pi_1(1-\pi_2)}{\pi_2(1-\pi_1)}}$$

Interpretation:

- $\psi = 1$: No association (same odds in both groups)
- $\psi > 1$: Group 1 has higher odds of success
- $\psi < 1$: Group 1 has lower odds of success

Estimating the Odds Ratio

Sample estimates:

$$\hat{\pi}_1 = \frac{y_1}{m_1}, \qquad \hat{\pi}_2 = \frac{y_2}{m_2}$$

Estimated odds ratio:

$$\hat{\psi} = \frac{\hat{\pi}_1}{\hat{\pi}_2} = \frac{y_1(m_2 - y_2)}{y_2(m_1 - y_1)}$$

Log odds ratio:

$$\log \hat{\psi} = \log y_1 - \log(m_1 - y_1) - \log y_2 + \log(m_2 - y_2)$$

Confidence Interval for Odds Ratio

Standard error of log odds ratio:

$$\mathrm{SE}(\log \hat{\psi}) = \sqrt{\frac{1}{y_1} + \frac{1}{m_1 - y_1} + \frac{1}{y_2} + \frac{1}{m_2 - y_2}}$$

95% CI for $\log \psi$:

$$\log \hat{\psi} \pm 1.96 \cdot \mathrm{SE}(\log \hat{\psi})$$

95% CI for ψ:

$$\left(\hat{\psi} \cdot e^{-1.96 \cdot \mathrm{SE}}, \quad \hat{\psi} \cdot e^{+1.96 \cdot \mathrm{SE}} \right)$$

If CI includes 1: No significant association.

Connection to Logistic Regression

Model: $\text{logit}(\pi_i) = \alpha + \beta \cdot \text{Group}_i$
where $\text{Group}_i = 0$ for group 2 (reference), $\text{Group}_i = 1$ for group 1.

Then:

$$\text{logit}(\pi_2) = \alpha$$
$$\text{logit}(\pi_1) = \alpha + \beta$$

Therefore:

$$\beta = \text{logit}(\pi_1) - \text{logit}(\pi_2) = \log \psi$$

And:

$$\boxed{e^{\beta} = \psi = \text{Odds Ratio}}$$

Example: Odds Ratio Calculation

Data:

	Disease	No Disease	Total
Exposed	30	70	100
Not Exposed	10	90	100

Odds ratio:

$$\hat{\psi} = \frac{30 \times 90}{10 \times 70} = \frac{2700}{700} = 3.86$$

Interpretation: Exposed individuals have 3.86 times the odds of disease compared to unexposed.

Relative Risk vs. Odds Ratio

Relative Risk (Risk Ratio):

$$\text{RR} = \frac{\pi_1}{\pi_2}$$

Odds Ratio:

$$\text{OR} = \frac{\pi_1/(1-\pi_1)}{\pi_2/(1-\pi_2)}$$

Relationship:

- When π_1, π_2 are small: $\text{OR} \approx \text{RR}$
- OR always more extreme than RR (farther from 1)
- OR has nicer mathematical properties
- OR is what logistic regression estimates

Summary: Overdispersion and Odds Ratios

Key points:

- Overdispersion: variance > binomial assumption
- Detect via $D/(n-r) \gg 1$
- Odds ratio: $\psi = O_1/O_2$
- $\psi = 1$ means no association
- From logistic regression: $\psi = e^{\beta}$
- CI for ψ via log transformation
- OR $\approx$ RR when probabilities are small

Next lecture: Dose-response experiments and applications.